

EXPERIMENT 4

1. **AIM:**

To perform Online Analytical Processing (OLAP) operations using Microsoft Power BI by creating multidimensional reports and applying Slice, Dice, Drill-down, and Drill-up operations on the Global Superstore dataset for analytical decision-making.

2. **OBJECTIVES:**

- To understand the basic concepts of OLAP in Business Intelligence.
- To perform Slice and Dice operations on business data.
- To perform Drill-down and Drill-up operations for detailed analysis.
- To analyze sales and profit across different dimensions.
- To create interactive reports using Power BI.
- To derive useful business insights from multidimensional data.

• **TOOLS / SOFTWARE REQUIRED:**

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor – built into Power BI Desktop
- DAX (Data Analysis Expressions)

3. **DATASET USED:** Global Superstore Dataset.xlsx.

4. **THEORY:**

OLAP (Online Analytical Processing) is a technique used in Business Intelligence to analyze large amounts of data from different perspectives. It helps users explore business information and identify trends, patterns, and relationships for decision-making.

Multidimensional Analysis allows business data to be viewed using different dimensions such as Time, Region, Category, Product, and Customer. Measures such as Sales, Profit, and Quantity can be analyzed across these dimensions.

Slice selects a particular value from one dimension to view a specific part of the dataset. For example, analyzing sales only for a particular year.

Dice selects multiple values from different dimensions to create a smaller subset of data. For example, analyzing sales for selected regions and product categories.

5. PROCEDURE:

Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.

Launch Microsoft Power BI Desktop.

Step 2: Import the Global Superstore dataset.

Home > Get Data > Excel Workbook

Select: Global Superstore Dataset.xlsx

Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:

The screenshot shows the Microsoft Power BI Desktop interface. The 'Navigator' pane on the left displays the 'Global Superstore 2018.xlsx' file with three worksheets selected: 'Orders', 'People', and 'Returns'. The main view shows the 'Returns' table with the following data:

Column1	Column2	Column3
Returned	Order ID	Region
Yes	IN-2017-CA120551-42816	Southern Asia
Yes	IN-2017-AA103751-42926	Southern Asia
Yes	IN-2017-TS212051-42904	Southern Asia
Yes	AG-2014-RO97803-41695	North Africa
Yes	AG-2015-LC70503-42265	North Africa
Yes	AG-2014-DB32103-41987	North Africa
Yes	AG-2017-AC4203-42796	North Africa
Yes	AG-2017-RD96603-43073	North Africa
Yes	AG-2015-CP23403-42152	North Africa
Yes	AG-2016-HF49953-42437	North Africa
Yes	AG-2015-CC24303-42247	North Africa
Yes	AG-2017-MO78003-42948	North Africa
Yes	AO-2017-BE14554-42954	Central Africa
Yes	AO-2017-GM45004-42996	Central Africa
Yes	AO-2016-IG50854-42668	Central Africa
Yes	US-2017-SR204255-42937	South America
Yes	US-2014-JP155205-41883	South America
Yes	MX-2016-LC171405-42457	South America
Yes	US-2016-JK152055-42585	South America
Yes	US-2014-JC153405-41897	South America
Yes	US-2017-CA119655-43061	South America
Yes	US-2016-AB1100455-42505	South America

The interface also shows the 'Visualizations' pane on the right with various chart options and the 'Data' pane at the bottom.

- Orders
- Returns
- People

Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables

Step 4: Analyse the Orders table.

Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity

- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.

The screenshot shows the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, showing options like 'Manage relationships', 'New measure', 'Quick measure', 'New table', and 'Mark as date table'. The 'Structure' pane on the left shows the 'Orders' table. The 'Data' pane on the right shows the 'Orders' table selected. The main area displays a table with 51,290 rows and 10 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, and Postal Code.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Co
27884	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
27883	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
26896	ID-2014-FP143207-41945	Sunday, November 2, 2014	Saturday, November 8, 2014	Standard Class	FP-143207	Frank Preis	Consumer	
30039	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30040	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30041	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
27444	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27443	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27445	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27446	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
25656	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25655	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25657	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25658	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
27561	ID-2015-CS124607-42189	Saturday, July 4, 2015	Wednesday, July 8, 2015	Standard Class	CS-124607	Chuck Sachs	Consumer	
28358	IN-2016-JF154907-42399	Saturday, January 30, 2016	Thursday, February 4, 2016	Standard Class	JF-154907	Jeremy Farry	Consumer	
24819	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
24818	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
22411	ID-2016-DJ136307-42524	Friday, June 3, 2016	Thursday, June 9, 2016	Standard Class	DJ-136307	Doug Jacobs	Consumer	
22118	IN-2016-LS172307-42490	Saturday, April 30, 2016	Friday, May 6, 2016	Standard Class	LS-172307	Lycoris Saunders	Consumer	

Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data.

Organize the attributes into:

A. Fact Table: Fact_Sales

B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

This classification forms the basis of the Star Schema.

Part C: Create Dimension Tables

Step 7: Open Power Query Editor.

Home > Transform Data

Step 8: Create Dim_Customer.

The screenshot shows the Power Query Editor with a table named 'Table.TransformColumns(#"Removed Blank Rows",{{"Customer ID", Text.Proper, type text}, {"Customer Name", Text.Proper, type text}, {"Customer Segment", Text.Proper, type text}})'. The table has 3 columns: Customer ID, Customer Name, and Customer Segment. The Customer ID column contains values like 'Ca-120551', 'Bd-116051', etc. The Customer Name column contains names like 'Cathy Armstrong', 'Brian Dahlen', etc. The Customer Segment column contains values like 'Home Office', 'Consumer', 'Corporate', etc. The Query Settings pane on the right shows the name 'Dim_Customer' and the applied steps: Source, Removed Other Columns, Removed Duplicates, Removed Blank Rows, and Capitalized Each Word.

Customer ID	Customer Name	Customer Segment
Ca-120551	Cathy Armstrong	Home Office
Bd-116051	Brian Dahlen	Consumer
Aj-107801	Anthony Jacobs	Corporate
Gm-144551	Gary Mitchum	Home Office
Vb-217451	Victoria Brennan	Corporate
Lo-171701	Lori Olson	Corporate
Ss-201401	Saphira Shifley	Corporate
Aa-103751	Allen Arnold	Consumer
Bg-110351	Barry Gonzalez	Consumer
Ah-105851	Angele Hood	Consumer
Cs-118451	Cari Sayre	Corporate
Dw-131951	David Wiener	Corporate
Rs-194201	Ricardo Sperren	Corporate
Ts-213401	Toby Swindell	Consumer
Jg-151151	Jack Garza	Consumer
Su-206651	Stephanie Ulpright	Home Office
DI-134951	Dionis Lloyd	Corporate
Eh-139451	Eric Hoffmann	Consumer
Ed-138851	Emily Ducich	Home Office
Jm-156551	Jim Mitchum	Corporate
Ts-212051	Thomas Seio	Corporate

Create a customer dimension containing:

- Customer ID
- Customer Name
- Segment

Remove duplicate records from the dimension.

Step 9: Create Dim_Product.

The screenshot shows the Power Query Editor with a table named 'Table.TransformColumns(#"Removed Blank Rows",{{"Product ID", Text.Proper, type text}, {"Product Name", Text.Proper, type text}, {"Product Sub-Category", Text.Proper, type text}, {"Product Category", Text.Proper, type text}})'. The table has 4 columns: Product ID, Product Name, Product Sub-Category, and Product Category. The Product ID column contains values like '861', '988', '1211', etc. The Product Name column contains names like 'Ikea Library With Doors, Mobile', 'Acme Scissors, Easy Grip', etc. The Product Sub-Category column contains values like 'Bookcases', 'Supplies', 'Machines', etc. The Product Category column contains values like 'Furniture', 'Office Supplies', 'Technology', etc. The Query Settings pane on the right shows the name 'Dim_Product' and the applied steps: Source, Removed Other Columns, Removed Duplicates, Removed Blank Rows, and Capitalized Each Word.

Product ID	Product Name	Product Sub-Category	Product Category
861	Ikea Library With Doors, Mobile	Bookcases	Furniture
988	Acme Scissors, Easy Grip	Supplies	Office Supplies
1211	Epson Receipt Printer, White	Machines	Technology
726	Rubbermaid Door Stop, Ergonomic	Furnishings	Furniture
564	Cameo Interoffice Envelope, With Clear Poly Window	Envelopes	Office Supplies
420	Bevis Conference Table, Fully Assembled	Tables	Furniture
626	Bush Classic Bookcase, Pine	Bookcases	Furniture
683	Hon Rocking Chair, Red	Chairs	Furniture
814	Samsung Audio Dock, Voip	Phones	Technology
129	Apple Audio Dock, Voip	Phones	Technology
780	Chromcraft Wood Table, With Bottom Storage	Tables	Furniture
747	Hoover Toaster, Red	Appliances	Office Supplies
588	Hewlett Personal Copier, High-Speed	Copiers	Technology
523	Harbour Creations Chairmat, Black	Chairs	Furniture
168	Smead Trays, Wire Frame	Storage	Office Supplies
758	Chromcraft Coffee Table, With Bottom Storage	Tables	Furniture
251	Motorola Headset, Cordless	Phones	Technology
08	Rogers Shelving, Wire Frame	Storage	Office Supplies
727	Rubbermaid Frame, Black	Furnishings	Furniture
812	Samsung Audio Dock, Cordless	Phones	Technology

Create a product dimension containing:

- Product ID
- Product Name
- Category
- Sub-Category

Remove duplicate records.

Step 10: Create Dim_Region.

Create a region dimension containing:

Power Query Editor - Query: Dim_Region

Formula Bar: `= Table.SelectRows("#Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems`

	City	State	Country	Region
1	Herat	Hirat	Afghanistan	Southern Asia
2	Kabul	Kabul	Afghanistan	Southern Asia
3	Kandahar	Kandahar	Afghanistan	Southern Asia
4	Jalalabad	Nangarhar	Afghanistan	Southern Asia
5	Durrës	Durrës	Albania	Southern Europe
6	Elbasan	Elbasan	Albania	Southern Europe
7	Korçë	Korçë	Albania	Southern Europe
8	Shkodër	Shkodër	Albania	Southern Europe
9	Vlorë	Vlorë	Albania	Southern Europe
10	Algiers	Algiers	Algeria	North Africa
11	Annaba	Annaba	Algeria	North Africa
12	Barika	Batna	Algeria	North Africa
13	Batna	Batna	Algeria	North Africa
14	Bechar	Bechar	Algeria	North Africa
15	Bejaia	Bejaia	Algeria	North Africa
16	Constantine	Constantine	Algeria	North Africa
17	Ain Oussera	Djelfa	Algeria	North Africa
18	Dar Chioukh	Djelfa	Algeria	North Africa
19	Messaad	Djelfa	Algeria	North Africa
20	Guelma	Guelma	Algeria	North Africa
21	Laghouat	Laghouat	Algeria	North Africa

4 COLUMNS, 199+ ROWS Column profiling based on top 1000 rows

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- Region
- Country
- State
- City

Remove duplicate records.

Step 11: Create Dim_Date.

Create a date dimension containing:

- Order Date
- Year
- Quarter
- Month

Ensure that the date-related fields are correctly formatted.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files

Queries [7]

People Orders Returns Dim_Customer Dim_Product Dim_Region Dim_Date

fx = Table.SelectRows(#"Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))

	Order Date	Year	Quarter	Month Name
1	3/22/2017	2017	1	March
2	9/1/2015	2015	3	September
3	4/19/2015	2015	2	April
4	8/1/2017	2017	3	August
5	12/11/2017	2017	4	December
6	9/24/2016	2016	3	September
7	12/16/2015	2015	4	December
8	7/10/2017	2017	3	July
9	9/28/2015	2015	3	September
10	11/27/2016	2016	4	November
11	1/18/2016	2016	1	January
12	11/30/2014	2014	4	November
13	6/5/2015	2015	2	June
14	8/16/2014	2014	3	August
15	12/23/2017	2017	4	December
16	10/24/2017	2017	4	October
17	9/17/2016	2016	3	September
18	7/29/2014	2014	3	July
19	9/2/2017	2017	3	September
20	12/12/2014	2014	4	December
21	9/23/2016	2016	3	September

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 3:55 PM

Step 12: Remove duplicate records.

Remove duplicate records from all dimension tables and verify that the key fields contain unique values.

Part D: Create Fact Table

Step 13: Create Fact_Sales.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows Transpose Reverse Rows Detect Data Type Fill Pivot Column Convert to List Replace Values Unpivot Columns Merge Columns Split Column Format Extract Parse Statistics Standard Scientific Rounding Information Date Time Duration Run R script Run Python script

Queries [8]

People Orders Returns Dim_Customer Dim_Product Dim_Region Dim_Date Fact_Sales

fx = Table.TransformColumns(Table.TransformColumnTypes(#"Removed Blank Rows", {"Order Date", type text}, {"Sales", type text}, {"Quantity", type text}), {"Order Date", type text}, {"Sales", type text}, {"Quantity", type text})

	Order ID	Order Date	Customer ID	Region	Product ID	Sales	Quantity
1	In-2017-Ca120551-42816	3/22/2017	Ca-120551	Southern Asia	Fur-Bo-4861	731.82	2
2	Id-2015-Bd116051-42248	9/1/2015	Bd-116051	Southern Asia	Off-Su-2988	243.54	9
3	In-2017-Ca120551-42816	3/22/2017	Ca-120551	Southern Asia	Tec-Ma-4211	346.32	3
4	In-2017-Ca120551-42816	3/22/2017	Ca-120551	Southern Asia	Fur-Fu-5726	169.68	4
5	Id-2015-Bd116051-42248	9/1/2015	Bd-116051	Southern Asia	Off-En-3664	203.88	4
6	Id-2015-Aj107801-42113	4/19/2015	Aj-107801	Southern Asia	Fur-Ta-3420	4626.1499999999996	5
7	In-2017-Gm144551-42948	8/1/2017	Gm-144551	Southern Asia	Fur-Bo-3626	2070.15	5
8	In-2017-Vb217451-43080	12/11/2017	Vb-217451	Southern Asia	Fur-Ch-4683	914.34	7
9	In-2016-Lo171701-42637	9/24/2016	Lo-171701	Southern Asia	Tec-Ph-5814	1168.44	7
10	In-2017-Vb217451-43080	12/11/2017	Vb-217451	Southern Asia	Tec-Ph-3129	500.94	3
11	Id-2015-Ss201401-42354	12/16/2015	Ss-201401	Southern Asia	Fur-Ta-3780	1447.56	3
12	In-2017-Aa103751-42926	7/10/2017	Aa-103751	Southern Asia	Off-Ap-4747	669.12	8
13	In-2015-Bg110351-42275	9/28/2015	Bg-110351	Southern Asia	Tec-Co-4588	427.41	3
14	In-2016-Ah105851-42701	11/27/2016	Ah-105851	Southern Asia	Fur-Ch-4523	417.42	6
15	In-2016-Cs118451-42387	1/18/2016	Cs-118451	Southern Asia	Off-St-6068	332.85	7
16	In-2016-Cs118451-42387	1/18/2016	Cs-118451	Southern Asia	Fur-Ta-3758	267.08999999999997	1
17	In-2015-Bg110351-42275	9/28/2015	Bg-110351	Southern Asia	Tec-Ph-5251	412.95	5
18	In-2017-Aa103751-42926	7/10/2017	Aa-103751	Southern Asia	Off-St-5708	244.8	4
19	In-2014-Ah105851-41973	11/30/2014	Ah-105851	Southern Asia	Fur-Fu-5727	219.78	2
20	In-2015-Dw131951-42160	6/5/2015	Dw-131951	Southern Asia	Tec-Ph-5812	848.4	5
21							

9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 3:55 PM

Create the central Fact Table containing:

- Order ID
- Customer ID
- Product ID
- Region
- Order Date
- Sales
- Quantity
- Profit
- Discount

Step 14: Verify key columns.

Verify that all key columns required to establish relationships with the Dimension Tables are present and correctly formatted.

The resulting Fact_Sales table acts as the central table of the Star Schema.

Part E: Build Star Schema

Step 15: Open Model View.

Select the Model View icon from the left navigation pane in Power BI.

Step 16: Create relationships.

Create relationships between:

The screenshot shows the 'Manage relationships' window in Power BI. It lists four relationships between the Fact_Sales table and dimension tables. Each relationship is a one-to-many relationship, indicated by the asterisk and 1 on the Fact_Sales side and the 1 on the dimension table side.

From: table (column)	Relationship	To: table (column)	Status
Fact_Sales (Customer ID)	*	Dim_Customers (Customer ID)	Active
Fact_Sales (Order Date)	*	Dim_Date (Order Date)	Active
Fact_Sales (Product ID)	*	Dim_Products (Product ID)	Active
Fact_Sales (Region)	*	Dim_Region (Region)	Active

The left pane shows the 'Dim_Customers' table with columns 'Customer ID' and 'Customer Name'. The right pane shows the 'Fact_Sales' table with columns 'Sales', 'Quantity', 'Profit', and 'Discount'.

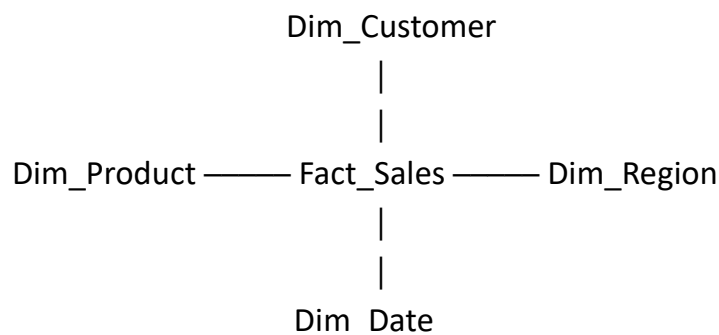
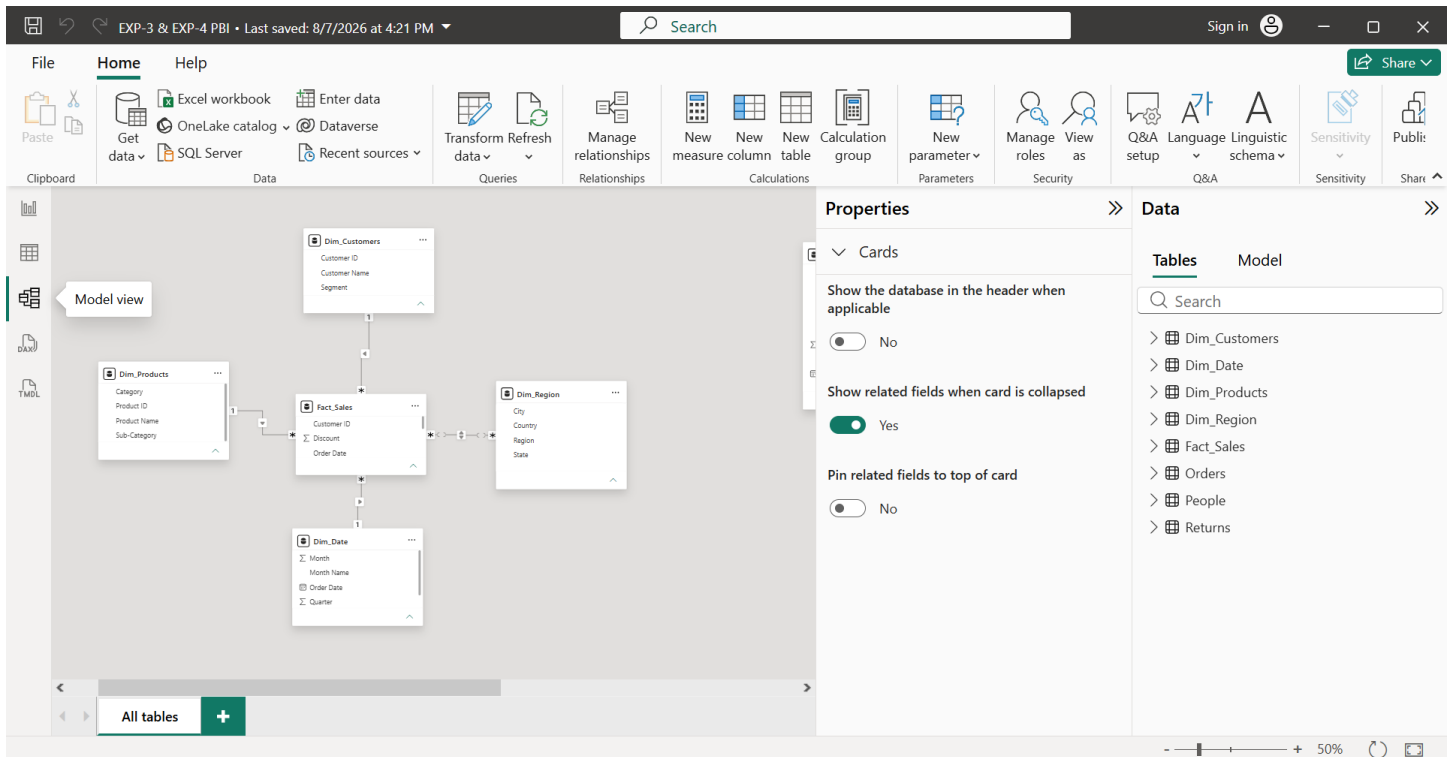
- Fact_Sales ↔ Dim_Customer
- Fact_Sales ↔ Dim_Product
- Fact_Sales ↔ Dim_Region
- Fact_Sales ↔ Dim_Date

Configure relationships.

Set the relationships appropriately as Many-to-One, with the Fact Table on the many side and the corresponding Dimension Table on the one side.

Step 18: Arrange the model.

Arrange the tables visually so that:



This forms the Star Schema with Fact_Sales at the centre and the Dimension Tables . surrounding it.

Part F: Validate Model and Create DAX Measures

Step 19: Create Power BI visualizations.

Create visualizations using the dimensional model to analyse the data.

Examples include:

- Sales by Category
- Profit by Region
- Sales Trend by Year

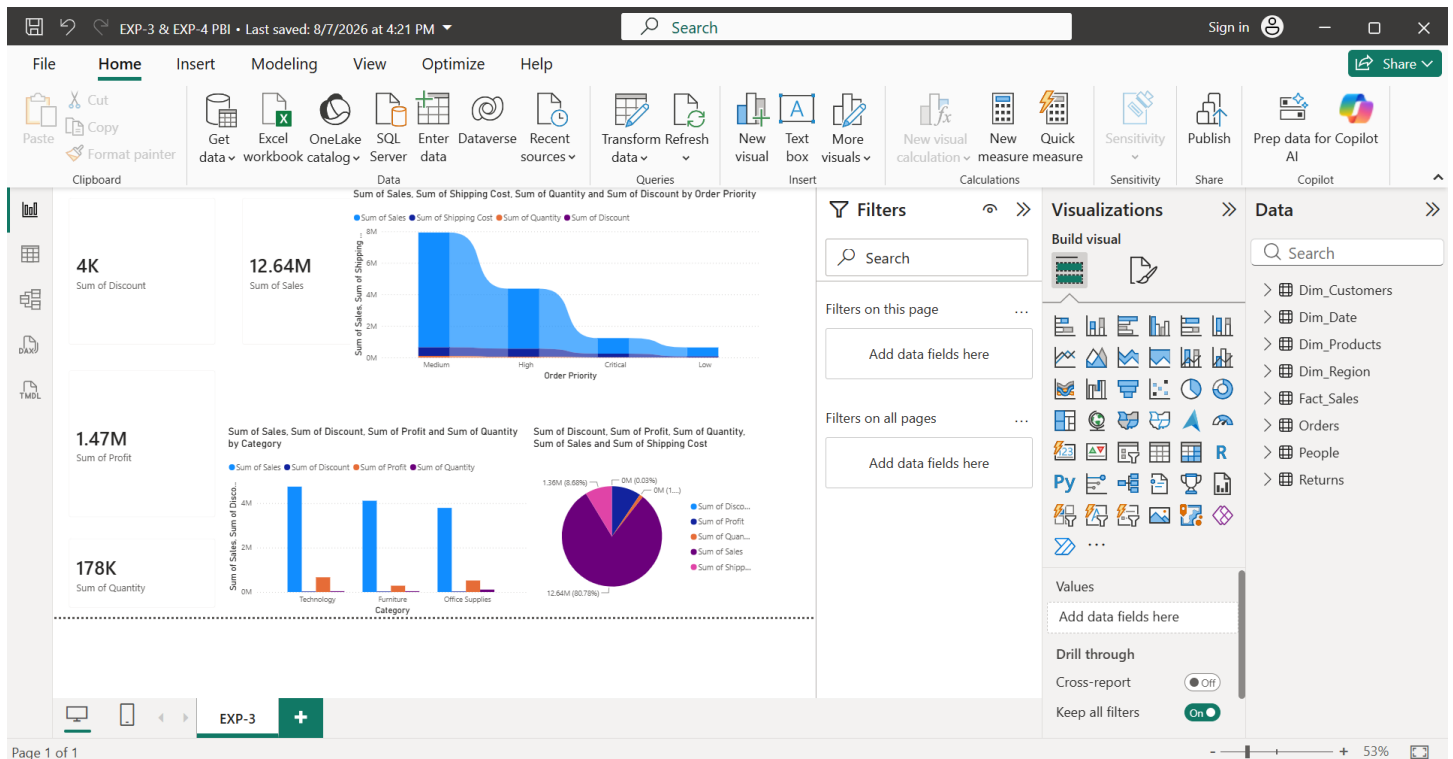
The reference specifies these as examples of visualizations for validating the model.

Step 20: Create DAX measures.

Create the following measures:

1. Total Sales = SUM(Fact_Sales[Sales])
2. Total Profit = SUM(Fact_Sales[Profit])
3. Profit Margin = DIVIDE(SUM(Fact_Sales[Profit]),SUM(Fact_Sales[Sales]))
4. Average Order Value = DIVIDE([Total Sales],DISTINCTCOUNT(Fact_Sales[Order ID]))

Step 21: Validate the Star Schema.

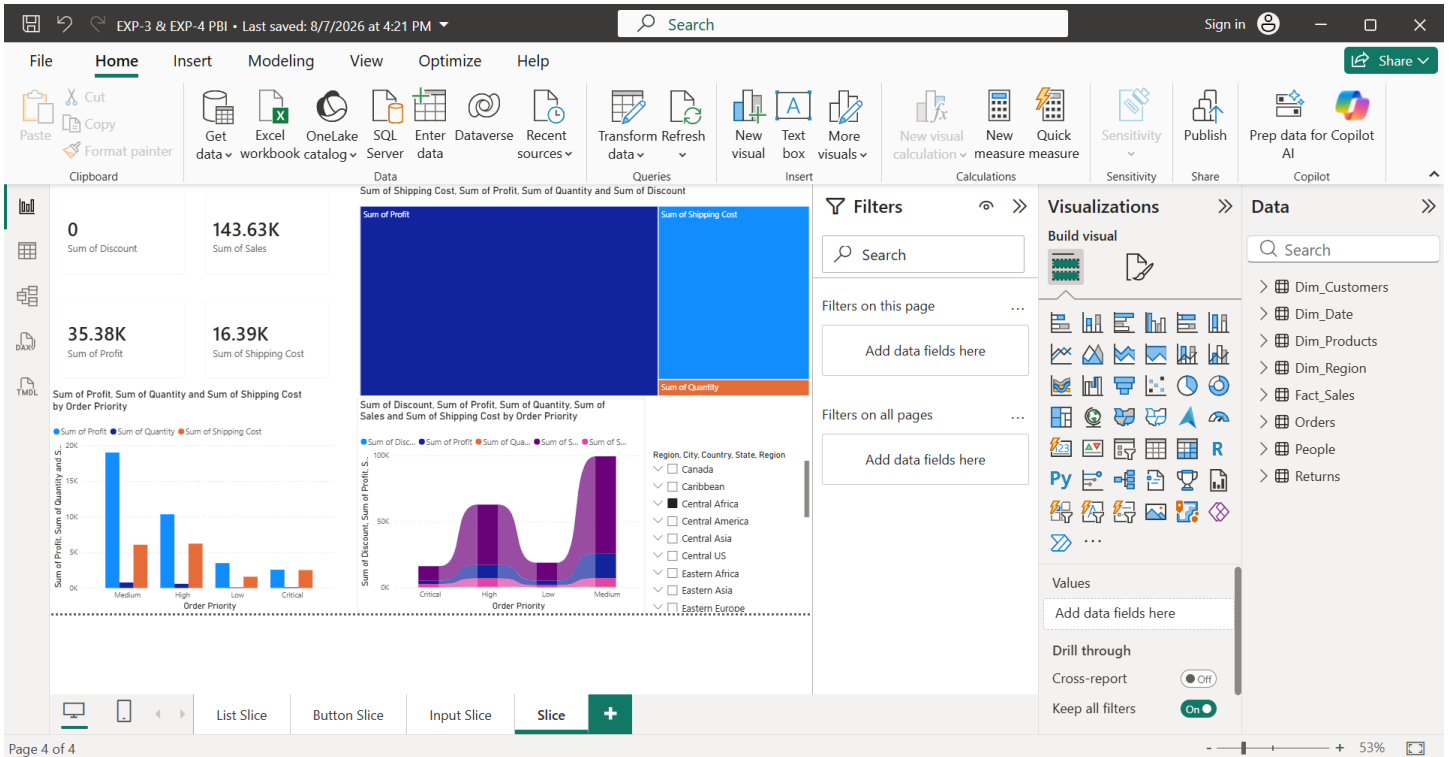


Use the created Dimension Tables to filter the Fact Table and verify that the relationships work correctly.

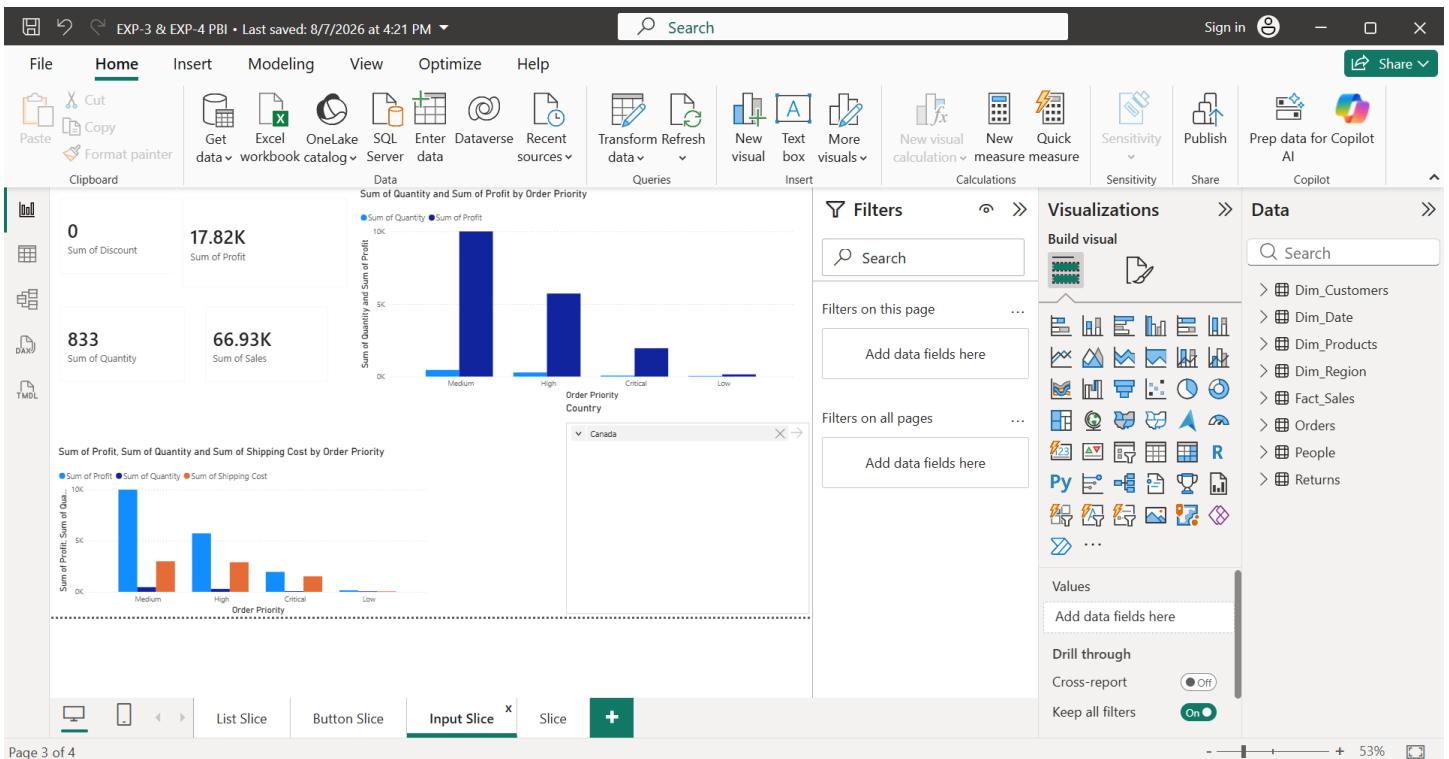
Check that changes in dimensions such as Customer, Product, Region, and Date correctly affect the corresponding Fact Table data and dashboard visualizations.

Conduct - Part G: Slice Operation

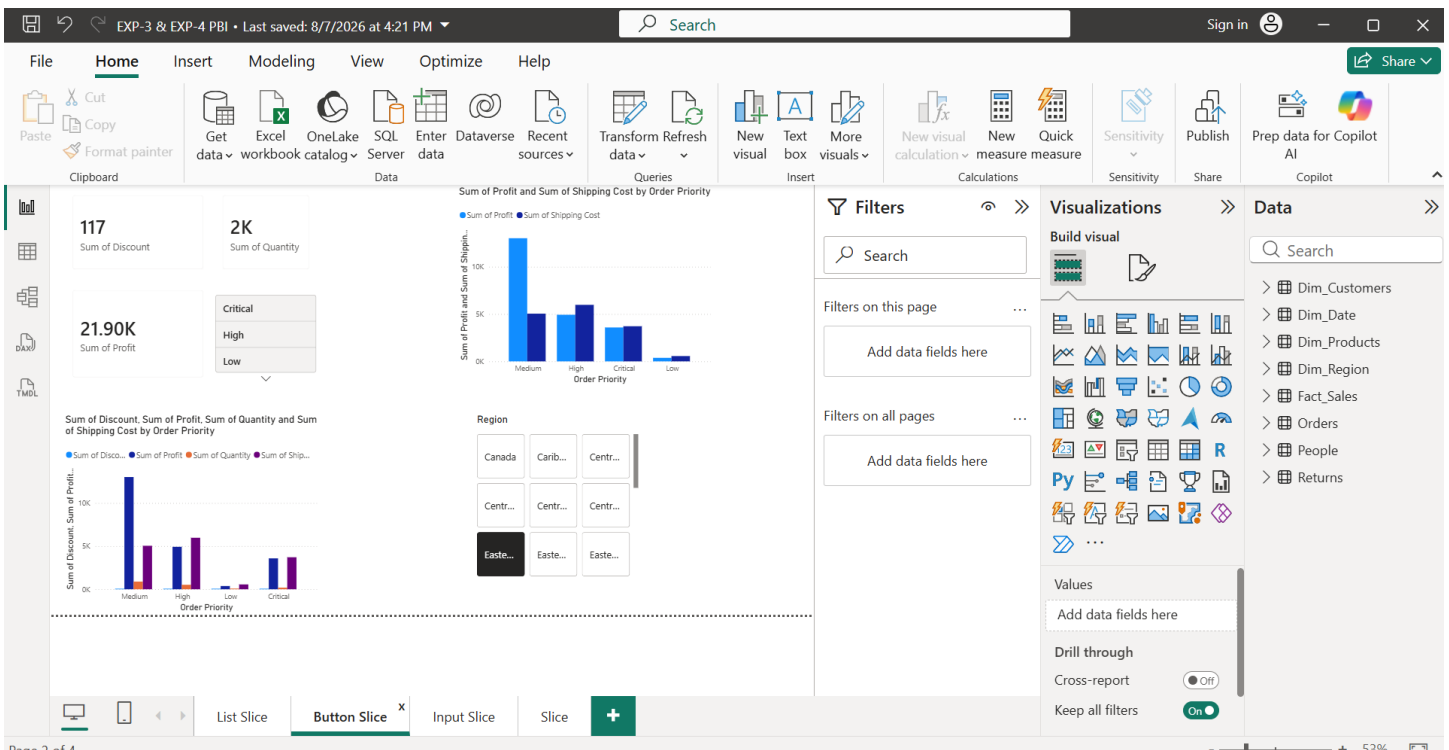
Step 22: Slice Operation



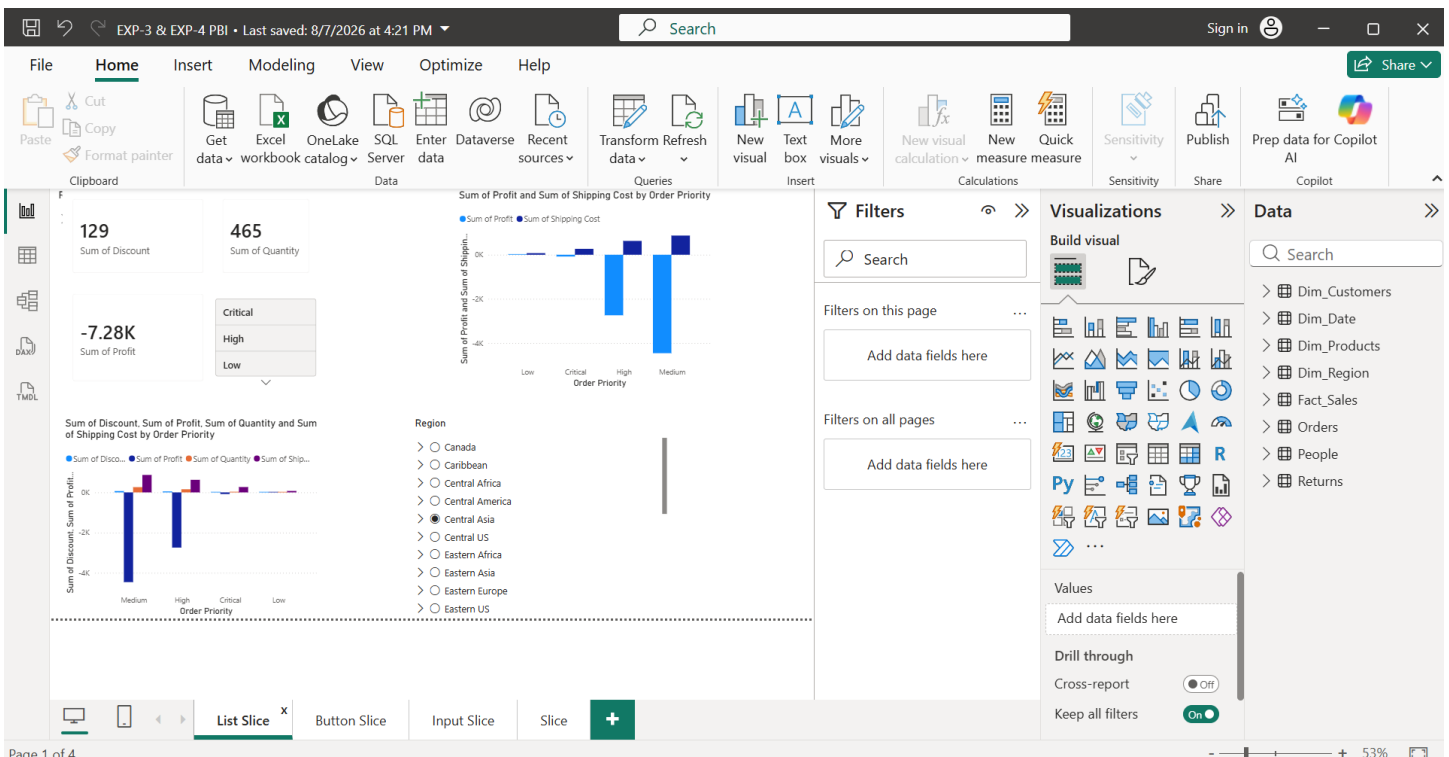
Step 23: Input Slice



Step 24: Button Slice



Step 25: List Slice



Conduct - Part H: Dice Operation

Step 26: Apply multiple filters simultaneously.

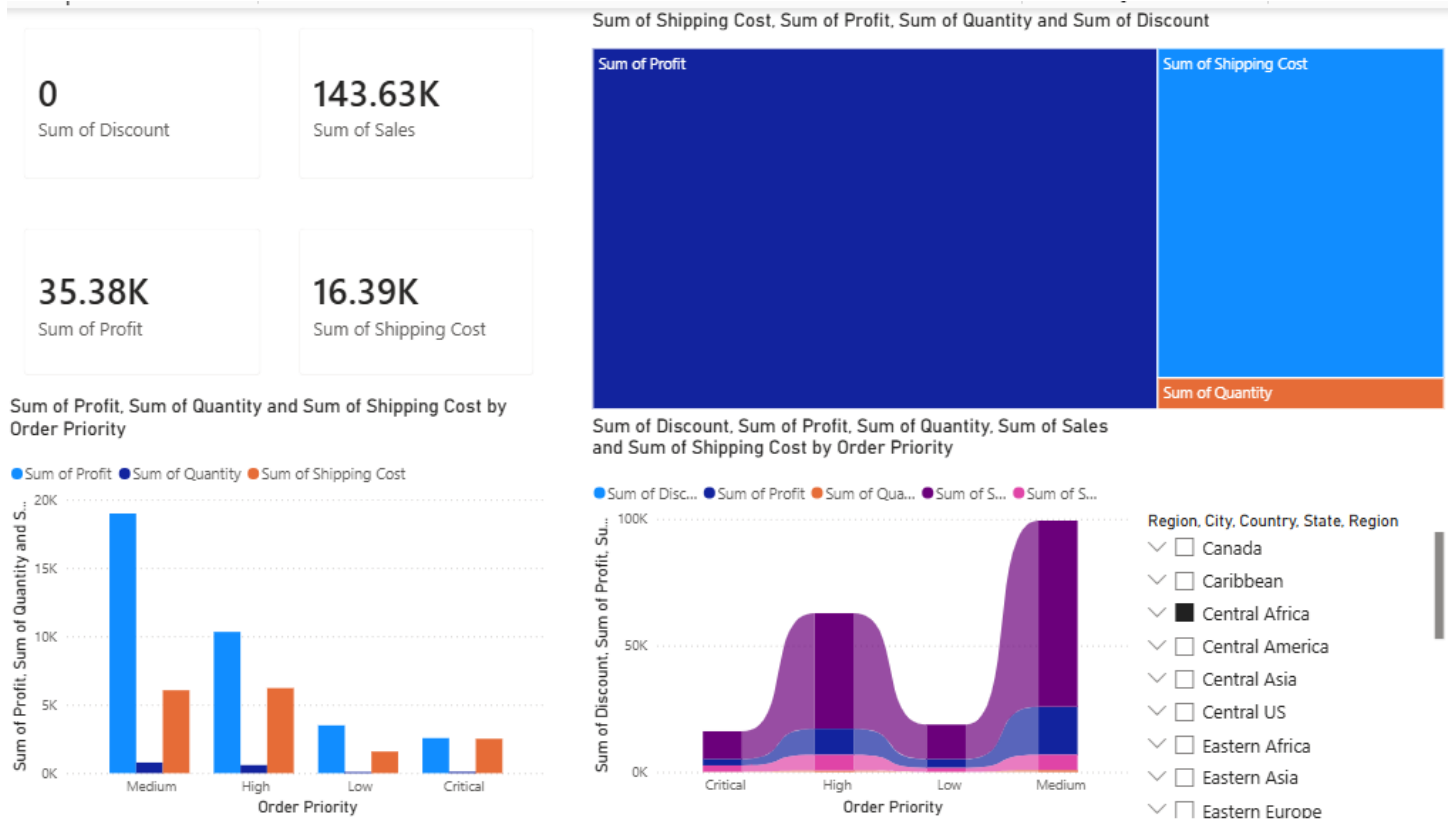
Example:

- Region = West
- Category = Technology

- Year = 2022

Step 27: Analyze filtered business metrics.

Step 28: Record observations and capture screenshots.



Conduct - Part I: Drill-down Operation

Step 29: Create a hierarchy using Region → State → City.

Step 30: Create a bar chart using the hierarchy.

Step 31: Use Drill-down mode to navigate from Region level to City level.

Step 32: Observe detailed sales patterns.

Step 33: Capture screenshots.

Conduct - Part J: Drill-up Operation

Step 34: Navigate back from City level to State level and then Region level.

Step 35: Compare aggregated values at each level.

Step 36: Record observations.

Conduct - Part K: OLAP Analysis

Step 37: Compare results obtained from Slice, Dice, Drill-down, and Drill-up operations.

Step 38: Create a summary table documenting observations.

Step 39: Explain how each operation supports business decision-making.



6. RESULT

OLAP operations were successfully performed on the Global Superstore dataset using Microsoft Power BI. Slice, Dice, Drill-down, and Drill-up operations were applied to analyze sales and profit across different dimensions, providing useful insights for business analysis.

7. LEARNING OUTCOMES

- Understood the concept and purpose of OLAP in Business Intelligence.
- Learned how multidimensional data can be analyzed from different perspectives.
- Gained practical experience with Slice and Dice operations.
- Learned how to navigate between summarized and detailed information using Drill-down and Drill-up.
- Learned how to use Power BI filters and hierarchies for analytical exploration.
- Understood how sales and profit can be compared across different dimensions.
- Gained experience in creating interactive OLAP-based reports.
- Learned how OLAP analysis can support business decision-making.